

Module 01: Systems — What Makes a Circle a Circle

Learning Objectives

1. Identify the Markov blanket of a crochet circle — who belongs, what resources are shared, and where the boundary lies between the group and the wider world.
2. Distinguish between the internal states, external states, and blanket states of a crochet circle as a social system.
3. Explain how a crochet circle maintains its identity over time even as members come and go.

Introduction

Picture a Tuesday evening. Five people gather around a table at the community center. Yarn bags thump down, hooks emerge, conversation picks up mid-sentence from last week. This is a crochet circle — and it is, in every formal sense, a **system**.

In Active Inference, a system is any entity that maintains a boundary between itself and its environment, preserving its own organization over time. A single crocheter is a system (as we explored in Course 1). But a crochet circle is a system at a higher scale — a social system, with its own internal states, its own relationship to the outside world, and its own boundary that defines who and what belongs.

This module asks a deceptively simple question: what makes a circle a circle? Not just any group of people crocheting, but *this* circle — the one that meets on Tuesdays, that has its own rhythms and running jokes, that somehow persists even when a founding member moves away or a newcomer joins for the first time.

Key Concepts

1. The Crochet Circle as a System with a Markov Blanket

A **Markov blanket** separates the internal states of a system from its external states. For a crochet circle, this boundary is social rather than physical, but it is no less real.

The **internal states** of a crochet circle include everything that makes it distinctly itself:

- **Shared knowledge:** the collective memory of techniques tried, patterns completed, mistakes learned from. "Remember when we all tried Tunisian crochet and nobody could figure out the return pass?" This accumulated experience is internal state — it exists within the circle and is not directly accessible to outsiders.
- **Ongoing projects:** the blanket someone has been working on for three months, the baby hat rush when a member announces a pregnancy, the group challenge to each make something with the same colorway.
- **Social bonds:** the trust that lets someone say "I frogged the whole thing" without embarrassment, the shorthand that develops ("pass me the good scissors"), the understanding of each person's pace and style.
- **Group norms:** the unwritten rules about when to offer help and when to let someone figure it out, whether the circle uses patterns or freehands, how new members are welcomed.

The **external states** include everything outside the circle that it exists in relation to:

- The wider crafting community (online forums, yarn shops, craft fairs)
- The local community (the venue that hosts meetings, neighbors who see the group)
- The broader culture of crochet (trends, new techniques, published patterns)
- Each member's life outside the circle (families, jobs, other hobbies)

The **blanket states** — the boundary itself — are the mechanisms through which inside and outside interact:

- **Sensory states** (how the outside affects the inside): a member brings a new pattern from a magazine, someone shares a technique they learned on YouTube, a yarn shop announces a sale that changes what materials the group works with.

- **Active states** (how the inside affects the outside): the circle hosts a craft fair booth, a member teaches a friend what they learned in the group, the circle's collective knowledge influences what the local yarn shop stocks.

2. Boundaries: Who Is In, Who Is Out

Every system requires a boundary, and crochet circles illustrate how social boundaries work. These boundaries are not walls — they are more like the selvedge edge of a woven fabric, defining the shape without being impermeable.

Regular members are clearly internal to the system. They carry the group's shared knowledge, understand its norms, and contribute to its ongoing projects. **Occasional visitors** sit at the boundary — they participate in the sensory and active exchanges of the meeting, but they do not carry the full internal state of the group. A visitor does not know that Maria always sits in the corner chair, or that the group tried entrelac last spring and decided it was not for them.

The meeting space is itself a blanket state. It is the physical location where internal and external states interact. When the circle meets at a coffee shop, the environment (noise level, table size, lighting) shapes the interaction. When they meet at someone's home, the intimacy of the space changes the quality of exchange. The space is not internal to the circle (they do not own it in most cases), but it is not fully external either — it is the medium through which the circle's life happens.

Shared materials also sit at the boundary. The communal stitch markers, the pattern books that live in someone's bag but are available to all, the extra hooks kept for newcomers — these objects mediate between individual and group.

3. Persistence Through Change

One of the most striking features of a system is its ability to maintain identity over time despite turnover in its components. A crochet circle demonstrates this vividly.

Members join and leave. The founding five may become three, then seven, then four. Someone moves away. A new person is invited by a friend. Yet the circle persists. It is still "the Tuesday circle." It still has its characteristic pace, its favorite techniques, its shared vocabulary.

In Active Inference, this persistence is understood through the concept of **self-evidencing**: a system acts in ways that confirm its own existence. The crochet circle self-evidences by continuing to meet, by maintaining its norms, by integrating new members into its existing culture rather than starting over each time someone new arrives.

When a long-time member leaves, the group experiences something like a perturbation — a disruption to internal states. The knowledge that person carried, the social role they filled, the projects they influenced — all of these must be redistributed or mourned. But the Markov blanket remains intact. The circle still meets, still stitches, still talks. The internal states reorganize around the loss.

When a new member joins, the boundary must accommodate them. The circle's sensory states register the newcomer — their skill level, their personality, their project interests. The circle's active states respond — someone explains the group's habits, offers help choosing a project, makes space at the table. Over time, the newcomer's knowledge and presence become part of the internal states, and the boundary shifts to include them.

Applications

- **The Lifecycle of a Circle:** Crochet circles are born, grow, and sometimes dissolve. A new circle begins when a few people decide to meet regularly — the Markov blanket is thin, the internal states are sparse. As the circle matures, internal states accumulate: shared history, inside jokes, collective skills. A circle dissolves when the boundary can no longer be maintained — too few members to sustain meetings, a venue lost, a conflict that fragments the group. Understanding this lifecycle through systems thinking helps us see what makes a circle resilient and what threatens it.
- **Nested Systems:** Every crochet circle is a system nested within larger systems. The circle exists within a local crafting community, which exists within a regional fiber arts culture, which exists within the global crochet world. Each level has its own Markov blanket. A trend that sweeps the online crochet world (like the viral crocheted granny square cardigan) enters the circle's external states through members who browse social media. It crosses the blanket when someone brings the pattern to a meeting. It becomes internal state when the group collectively decides to try it.

Conclusion

A crochet circle is far more than a collection of people who happen to crochet in the same room. It is a system — a self-maintaining entity with clear (if permeable) boundaries, rich internal states, and dynamic exchanges with the world beyond its edge. The Markov blanket of the circle is woven from meeting times, shared space, communal knowledge, and the social bonds that hold it all together. Understanding the circle as a system helps us appreciate why some circles thrive for decades while others dissolve after a few meetings, and why the simple act of gathering to stitch together creates something greater than the sum of its parts. In the next module, we turn our attention inward — to each individual maker sitting at the table.